Smart Clinic AI Assistant

Project Info Form

DEPI — Digital Egypt Pioneers Initiative | Track: AI & Data Science

---

Project name: Smart Clinic AI Assistant

Project Idea: An n8n-based automation system for a single-doctor clinic that handles patient communication end-to-end via a Telegram chatbot. All routing between general inquiries and appointment booking is done through Telegram inline-button menus (no LLM involved in that step), with real-time Google Calendar availability checks to prevent double-booking. The system also sends 24-hour interactive appointment reminders, generates an automated daily financial report (calculated in code, not by the LLM), and collects post-visit patient feedback, which an LLM classifies by sentiment — the only step in the system that uses an LLM. This reduces manual staff work and gives the doctor accurate, up-to-date visibility into clinic performance.

Github repo URL: https://github.com/nhahub/NHA-5-201

Tools, Libraries and APIs

- n8n (core automation/orchestration engine)
- Telegram Bot API
- Google Calendar API
- Google Sheets API
- Gmail API
- LLM API (feedback sentiment classification only)

Project Team Members

- Ahmed Hossam Abdullah
- Mohamed Alaa Ibrahim
- Ziad Ahmed Fathy
- Ahmed Mossa Anwar

Team Leader

- Ahmed Khaled Abdelfattah

Final Project Deliverable: A working n8n automation system (not a standalone app) — a set of deployed, interconnected n8n workflows exposed to patients through a Telegram bot, demoed live rather than packaged as an api/mobile/web app.

Project Pipeline:

1 Patient sends a message via Telegram.

2. Button menu routes the patient to Inquiry or Booking (no LLM).
3. Booking path checks Google Calendar for availability (code-based, no double-booking) before confirming — single doctor, 20-minute slots.
4. Confirmed data (patient, appointment, chat_id) is logged to Google Sheets — the shared data store.
5. From Sheets, three scheduled jobs run independently: Reminders (sent 24h before each appointment); Financial report (daily revenue and case count, calculated in code and emailed to the doctor); Feedback (LLM sentiment analysis on post-visit patient responses).

(Recreate this flow as a diagram in diagrams.net per the submission requirement — the stages above are the exact content to lay out.)

Dataset source: No fixed external dataset — the system operates on live data generated during operation (bookings, appointments, and feedback captured through Telegram/Calendar/Sheets), not pre-collected training data.

Timeline Deliverables

Week 1 & 2:

Weeks deliverables (project-wise):

- Finalize Project Planning & Management docs (Proposal, Plan, Task Assignment, Risk Assessment, KPIs).
- Finalize Requirements Gathering (Stakeholder Analysis, User Stories, Functional/Non-functional Requirements).
- Confirm GitHub repo setup and upload all finalized documents.
- Begin System Analysis & Design artifacts (Use Case Diagram, initial ER Diagram).
- Initial version of the Telegram booking bot is working (built by Mohamed) — covers message handling and Google Calendar availability checks.

Team Members' Roles:

- Ahmed Hossam Abdullah role: n8n automation & integrations (Modules 2–3 — booking and reminders)
- Mohamed Alaa Ibrahim role: Telegram bot logic and workflow build (Modules 1, 4–5 — messaging, financial report, feedback)
- Ziad Ahmed Fathy role: Testing & QA
- Ahmed Mossa Anwar role: Documentation & presentation support